

HARNETT REGIONAL, NC, USA

WMO: 746936

Lat: 35.379N

Lon: 78.734W

Elev: 60

StdP: 100.60

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 03712

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.2	-3.0	-18.8	0.7	1.4	-15.2	1.0	2.1	8.2	13.9	7.4	12.1	1.3	320	0.366

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	36.1	24.3	34.5	24.1	32.9	23.6	26.8	31.8	26.1	30.8	25.5	29.9	2.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.9	21.4	28.4	25.0	20.2	27.8	24.1	19.2	27.0	84.3	31.7	81.1	30.7	78.0	29.7	32.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.0	5.7	5.0	DB	-8.5	37.8	2.2	1.8	-10.1	39.1	-11.4	40.2	-12.6	41.3	-14.2	42.6	(4)
			WB	-9.8	28.2	2.1	1.4	-11.3	29.2	-12.6	30.0	-13.7	30.8	-15.3	31.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.2	6.6	8.2	12.1	17.5	21.9	25.7	27.3	26.8	23.1	17.3	11.4	8.4
	DBStd	8.52	5.61	5.17	5.26	4.56	3.71	2.91	1.99	2.43	3.12	4.53	4.66	5.11
	HDD10.0	404	137	89	40	3	0	0	0	0	0	3	37	94
	HDD18.3	1541	366	287	203	68	13	0	0	0	4	75	213	311
	CDD10.0	3042	31	37	104	229	368	472	536	519	394	228	79	44
	CDD18.3	1136	1	2	9	44	122	222	278	261	147	42	5	2
	CDH23.3	10613	7	15	92	441	1112	2194	2772	2513	1126	299	36	7
	CDH26.7	4383	1	1	12	126	404	983	1281	1118	391	63	4	0
Wind	WSAvg	1.7	1.9	2.0	2.3	2.2	1.8	1.6	1.5	1.3	1.6	1.4	1.5	1.6
Precipitation	PrecAvg	1198	94	73	99	81	96	114	148	133	120	83	80	77
	PrecMax	1559	200	161	180	196	162	344	244	274	450	195	211	176
	PrecMin	850	28	25	22	26	29	36	42	36	9	0	11	22
	PrecStd	170	45	36	41	43	40	78	58	51	96	48	49	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.8	25.6	27.9	32.3	35.0	37.7	37.3	37.9	35.0	30.8	27.3	24.4
		MCWB	18.3	18.0	17.6	20.8	22.5	24.4	24.4	24.3	22.0	22.8	19.2	19.7
	2%	DB	20.9	22.5	26.0	29.9	32.4	35.3	35.9	35.2	32.3	28.2	23.9	22.0
		MCWB	16.7	15.7	16.8	19.1	21.6	24.0	24.8	24.3	23.0	20.8	18.6	18.1
	5%	DB	17.9	19.8	23.8	27.8	30.9	33.7	34.0	33.6	30.9	26.4	22.1	19.1
		MCWB	14.4	15.0	15.7	18.5	21.3	23.2	24.8	24.6	22.1	19.8	16.7	16.6
	10%	DB	16.0	17.5	21.3	26.1	28.9	32.0	32.6	32.2	28.8	24.9	19.2	17.3
		MCWB	12.2	12.4	15.0	17.6	20.5	22.8	24.0	24.1	21.9	19.6	15.8	14.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.8	19.9	21.1	24.2	26.2	27.1	27.8	27.9	26.1	24.9	22.1	21.1
		MCDB	22.0	22.2	24.8	28.3	29.7	32.9	33.5	32.7	30.1	26.4	24.7	22.2
	2%	WB	17.7	17.8	19.2	22.4	24.5	25.9	26.8	26.8	25.0	23.7	19.9	19.1
		MCDB	19.2	20.8	22.3	26.2	28.3	31.2	32.2	31.2	28.9	26.2	21.7	20.8
	5%	WB	15.7	15.9	17.9	20.9	23.2	25.1	26.1	25.9	24.2	22.3	18.2	17.3
		MCDB	17.8	19.0	20.9	24.8	27.2	29.7	31.0	30.2	27.3	24.9	20.4	18.7
	10%	WB	13.1	13.3	16.7	19.4	22.1	24.3	25.4	25.4	23.5	20.9	16.4	14.9
		MCDB	15.0	16.5	20.1	23.4	26.4	28.8	30.1	29.5	26.4	23.5	19.2	16.9
Mean Daily Temperature Range	5% DB	MDBR	10.6	11.4	12.3	12.8	11.6	10.7	10.3	9.9	10.1	11.4	12.0	10.7
		MCDBR	13.5	14.5	14.7	14.9	13.6	12.8	11.9	11.9	12.1	12.6	13.6	12.3
		MCWBR	9.6	9.1	7.3	6.2	4.7	4.1	3.9	3.7	4.2	6.2	8.3	8.9
	5% WB	MCDBR	11.7	12.8	11.7	11.8	11.3	10.6	10.6	9.9	8.9	9.0	10.8	10.8
		MCWBR	9.8	9.9	8.1	6.9	5.5	4.2	4.2	3.8	4.2	5.4	8.4	9.2
Clear-Sky Solar Irradiance	taub	0.322	0.338	0.371	0.413	0.467	0.509	0.555	0.534	0.432	0.388	0.341	0.322	
	taud	2.472	2.422	2.363	2.245	2.136	2.062	1.951	1.996	2.280	2.373	2.464	2.509	
	Ebn at Noon	879	903	898	872	827	789	749	756	825	834	847	857	
	Edh at Noon	85	100	116	137	155	166	185	174	125	104	86	78	
All-Sky Solar Radiation	RadAvg	2.62	3.29	4.35	5.47	6.03	6.40	6.08	5.48	4.61	3.75	2.88	2.28	
	RadStd	0.17	0.38	0.31	0.37	0.46	0.43	0.37	0.28	0.38	0.48	0.23	0.19	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (58 neighbors)	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+160	+113

Nomenclature: See separate page